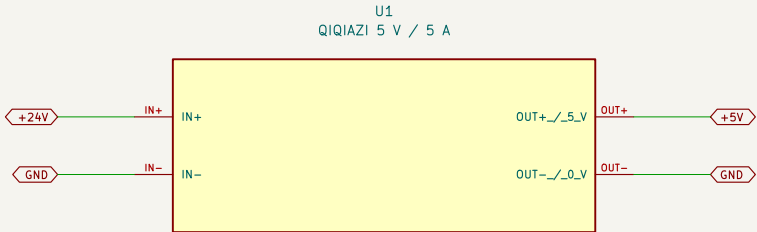
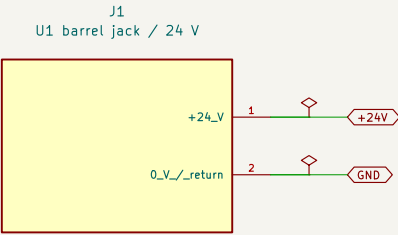


CONTROLLER AND POWER



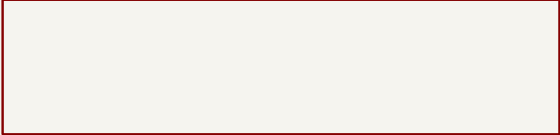
J1 is the existing barrel jack on buck U1.  
J103 VIN pins feed J112 and both TMC power headers.  
Dedicated underside supply/return branches; common 0 V.  
5 A is the converter rating, not the Nucleo input rating.

NUCLEO POWER CONFIGURATION  
JP2: fit EXT jumper on pins 5–6.  
Buck +5 V -> CN11 pin 6; ground -> CN11 pin 8.  
CN8 pin 7 supplies 3.3 V to interface/storage.  
SV\_EXT input: 4.75–5.25 V, 500 mA maximum.  
Power EXT first, then connect ST–Link USB CN1.  
Do not connect 24 V to Nucleo VIN or logic pins.  
STLINK–V3 MCO: HSE/5 = 5 MHz (persistent setting).  
Target HSE bypass; CPU 480 MHz / timer kernels 240 MHz.

BOARD ROUTING / TIM2 DMA STEP OUTPUTS  
PA0 -> CN10.29: SB75 ON; user button stays PC13.  
QSPI: SB59/SB63 ON; SB61/SB69 OFF.  
PE2 -> CN10.25: SB67 OFF.  
Only used external contacts are drawn.  
Ethernet, ST–Link, LEDs and button stay onboard.

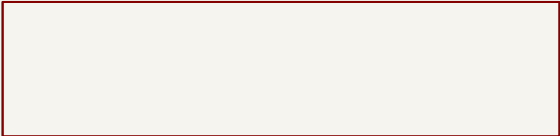
Shared names connect across sheets. A1 contacts use CNxx.pin numbering. J101–J115 are perfboard connectors.  
REF inputs: internal 3.3 V pull–ups, both–edge EXTI; NC to GND. Released = LOW; pressed / unplugged = HIGH.  
U1/A3/A4 mount at J103/J105/J106. Buck VIN distributes 24 V. TMC JP1.6/5: power headers; JP1.4–1: motor terminals.

M1 – DM542T



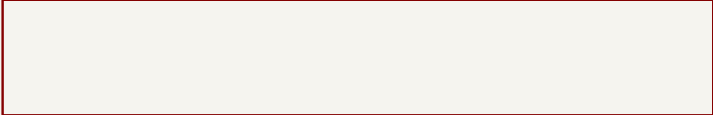
Date: sheets/M1\_DM542T.kicad\_sch

M1 – encoder interface



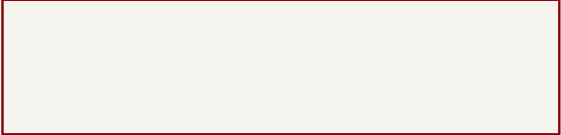
Date: sheets/M1\_Encoder.kicad\_sch

Perfboard cable headers



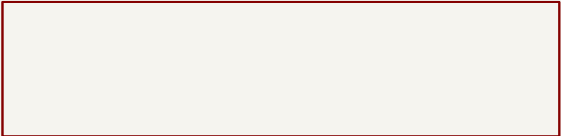
Date: sheets/Perfboard\_Connectors.kicad\_sch

M2 and M3 – TMC2209



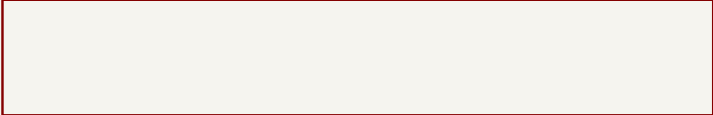
Date: sheets/M2\_M3\_TMC2209.kicad\_sch

QSPI flash and SD card



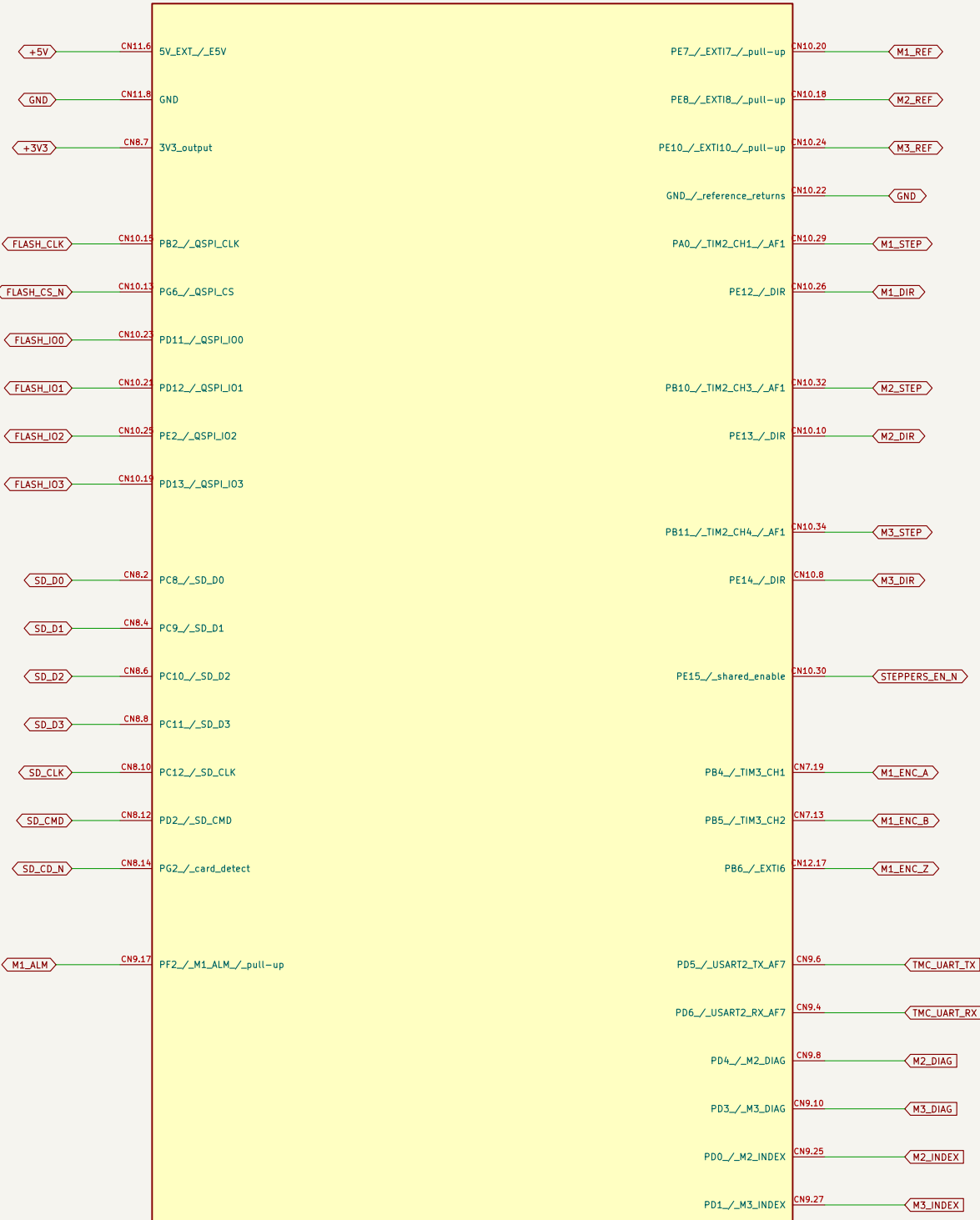
Date: sheets/Storage.kicad\_sch

Reference limit switches



Date: sheets/Reference\_Switches.kicad\_sch

A1  
NUCLEO–H753ZI / MB1364



System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /  
File: ProtoMate.kicad\_sch

Title: ProtoMate SCARA – Controller and power

Size: A3 Date: 2026–10–03

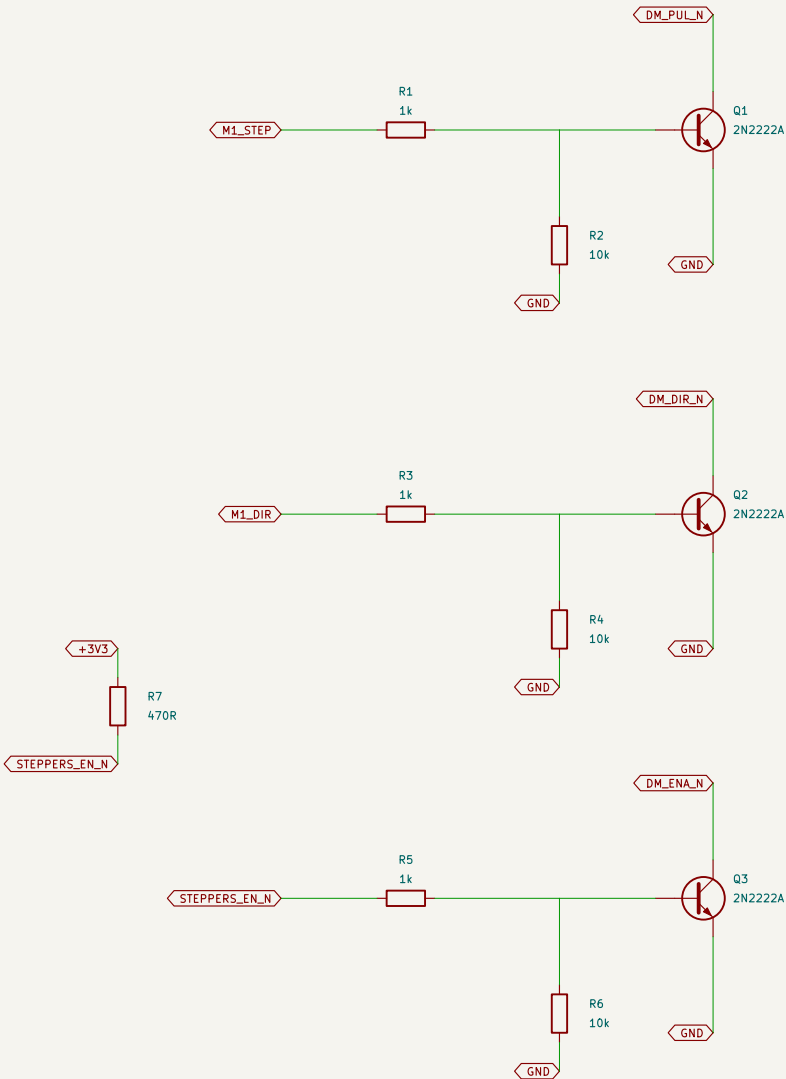
KiCad E.D.A. 10.0.6

Rev: 5.0

Id: 1/7

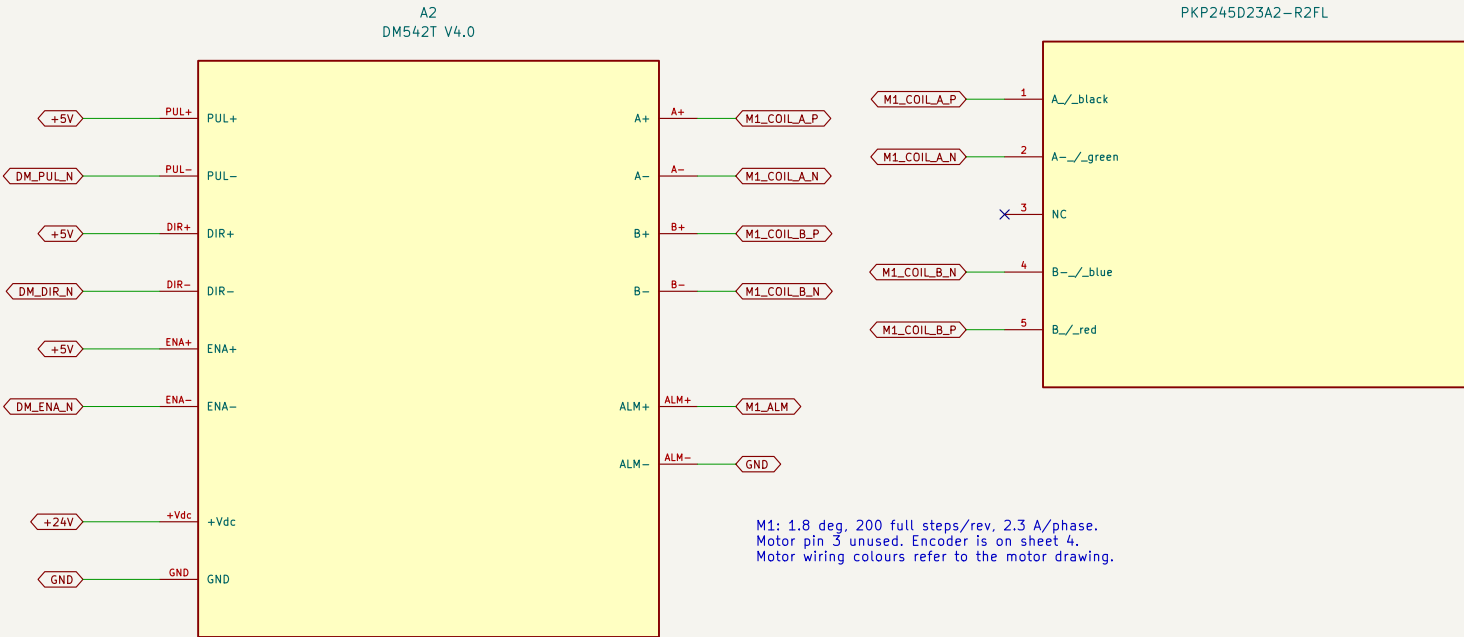
M1 – DM542T

3.3 V MCU -> 2N2222A current sinks -> 5 V optoisolated inputs



SHARED ENABLE AND TIMING  
PE15 low: Q3 off, DM542T enabled, both TMC EN inputs low.  
PE15 high: Q3 on, DM542T disabled, both TMC EN inputs high.  
R7 pulls PE15 high during MCU reset; about 7 mA sink when enabled.  
After enable, wait >=200 ms. DIR setup >=5 us; STEP high/low >=2.5 us.

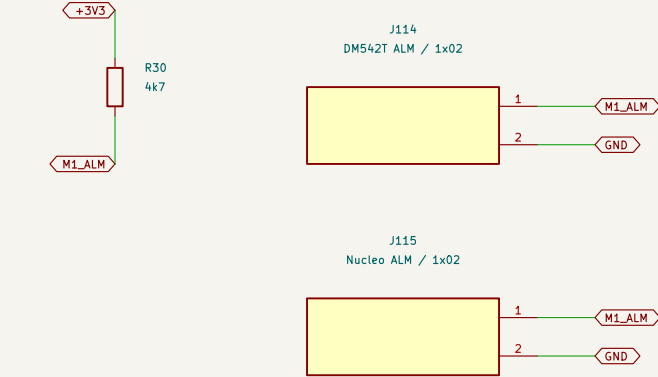
ALARM; ALM+ -> J114.1; ALM- -> J114.2 / GND.  
J115.1 -> PF2 / CN9.17; J115.2 -> Nucleo GND.  
R30 pulls ALM+ to 3.3 V (about 0.7 mA).  
Default ALM; conducts healthy, opens on fault.  
HIGH / unplugged: latch fault; disable all drivers.  
No 5 V / 24 V connection to ALM or PF2.



24 V power cable: J112.1 -> +Vdc; J112.2 -> GND.  
Signal cable: J104. Motor winding cable stays on A2.

DM542T SETTINGS / INITIAL COMMISSIONING  
S2 signal selector: 5 V.  
1/16 step: SW5 ON, SW6 ON, SW7 OFF, SW8 ON.  
3200 STEP pulses per motor revolution.  
Conservative initial current: 2.03 A RMS / 2.84 A peak  
SW1 ON, SW2 ON, SW3 OFF; SW4 OFF (idle reduction).  
Check the actual V4.0 label before setting switches.

Q1-Q3 show functional B/C/E pins, not package lead numbers.  
Verify your 2N2222A manufacturer/package before assembly.  
Fit each 10k between base and emitter. Keep PUL+/PUL- paired;  
separate signal cables from motor power cables.



System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

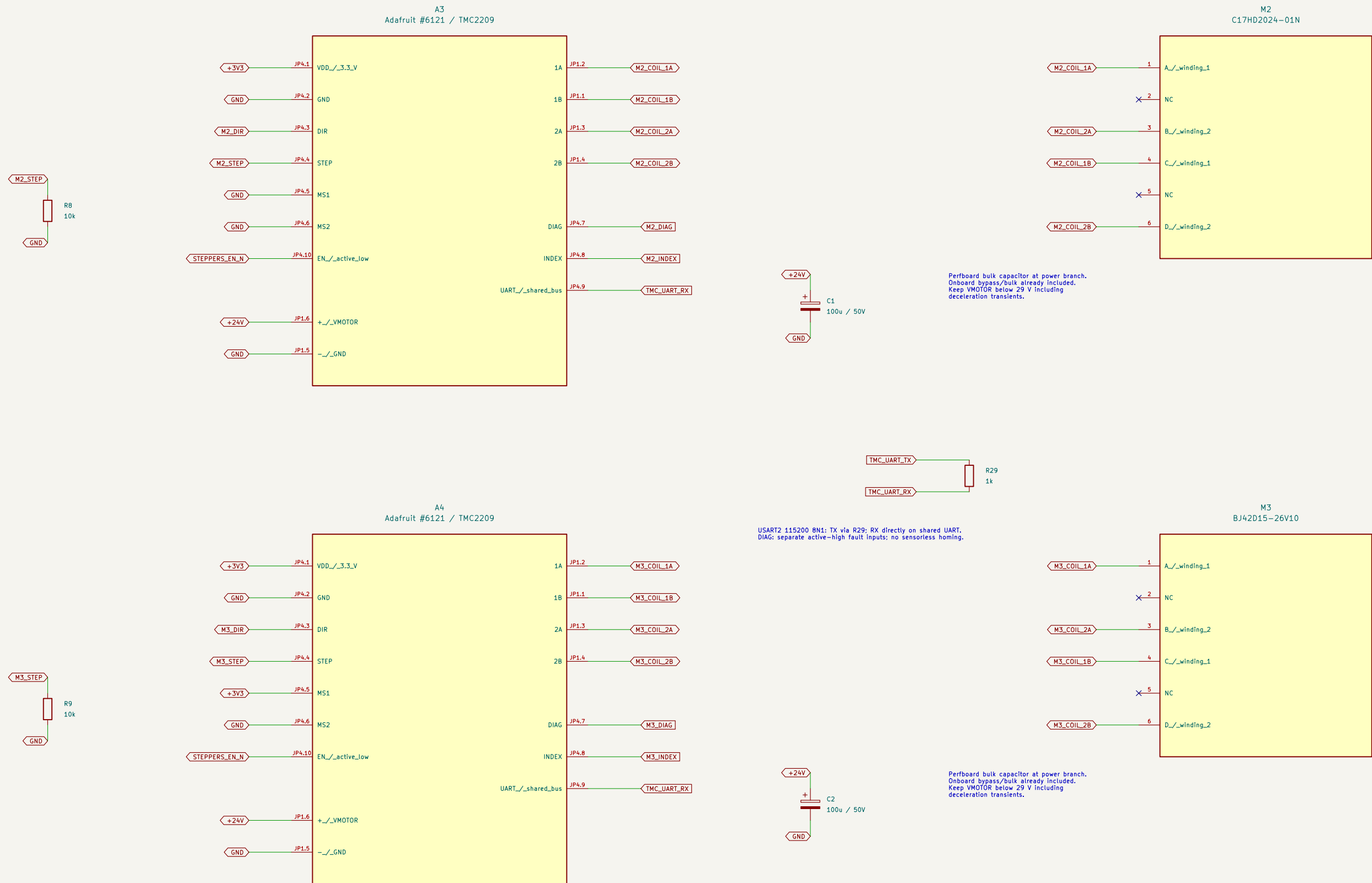
Sheet: /M1 – DM542T/  
File: M1\_DM542T.kicad\_sch

Title: ProtoMate SCARA – M1 – DM542T

|                     |                  |          |
|---------------------|------------------|----------|
| Size: A3            | Date: 2026-10-02 | Rev: 1.0 |
| KiCad E.D.A. 10.0.6 |                  | Id: 2/7  |

## M2 AND M3 – TMC2209

UART addresses: M2 MS1/MS2 = GND/GND (0); M3 = 3.3 V/GND (1). SPRD jumper OPEN.  
Firmware selects 1/16 step and SpreadCycle; digital current bypasses the potentiometers.



M2: 1.0 A/phase; UART run/hold 650 mA RMS requested (625 mA quantized).  
M3: 0.84 A/phase; UART run/hold 550 mA RMS requested (511 mA quantized).  
24 V motor supply; 3.3 V logic/UART. Drivers remain disabled until both answer and verify.  
A3/A4: JP4 + JP1.6/5 solder into J105/J106. JP1.4-1: four-way motor terminals.

System wiring and linked perfboard assembly map | all voltages DC

### ProtoMate Control

Sheet: /M2 and M3 – TMC2209/  
File: M2\_M3\_TMC2209.kicad\_sch

Title: ProtoMate SCARA – M2 and M3 – TMC2209

|          |                  |
|----------|------------------|
| Size: A3 | Date: 2026-10-02 |
|----------|------------------|

|                     |   |
|---------------------|---|
| Size: A5            | D |
| KiCad E.D.A. 10.0.6 |   |

Date: 2026-10-02

Rev: 1.0

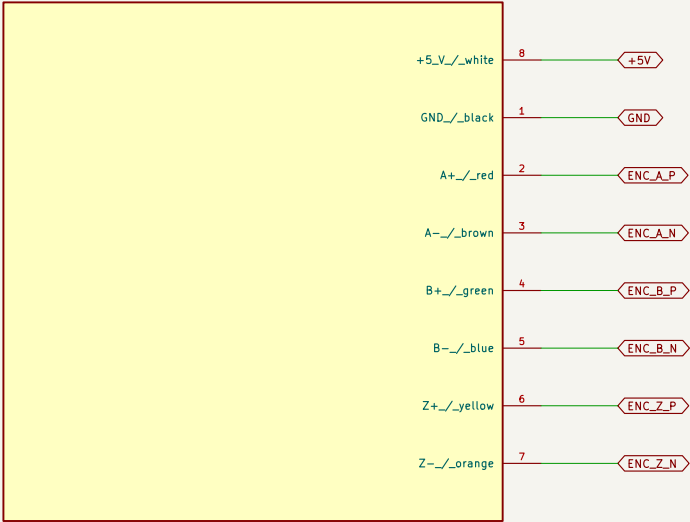
id: 3/7

M1 – ENCODER / HC126 + NPN LEVEL SHIFTERS

U2 input pairs are deliberately reversed.  
Q4–Q6 cancel that inversion before U3.  
Encoder connector pinout stays unchanged.  
Never wire U2 outputs directly to U3.

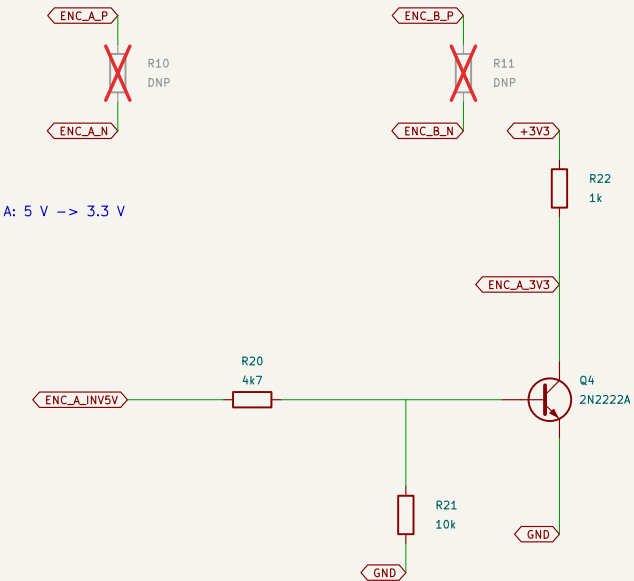
U2: 5 V. Q4–Q6 translate to 3.3 V. U3: 3.3 V ONLY.  
U3 OE1/OE2/OE3 HIGH; unused OE4 and A4 LOW.  
U2 unused receiver inputs may float (fail–safe).  
100 nF at each IC; 1 uF nearby.

J2  
M1 encoder / 8–pin



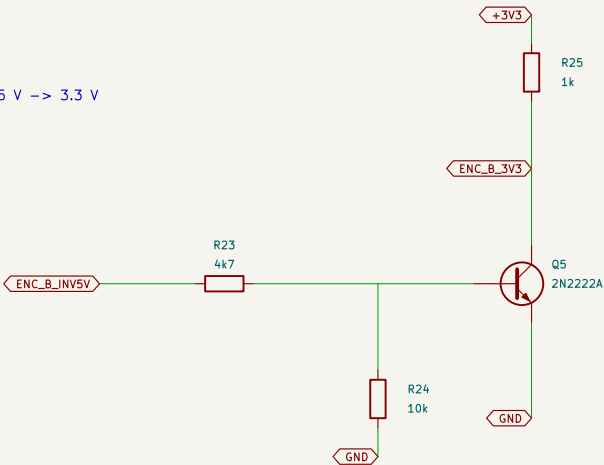
400 P/R -> TIM3 x4 = 1600 counts/rev; Z = once/rev.  
Cable colours / connector orientation; motor manual.  
Shielded twisted pairs; chassis bond at cable entry.

R10–R12: OPTIONAL TERMINATION, DNP.  
Choose for cable and 20 mA encoder output limit.  
A+/A-, B+/B-, Z+/Z-: twisted pairs.

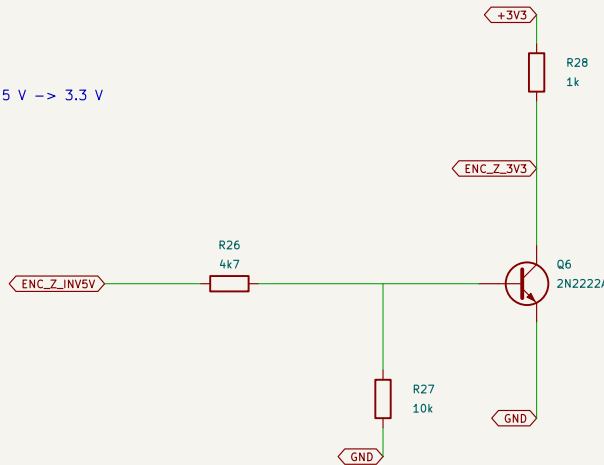


A: 5 V -> 3.3 V

B: 5 V -> 3.3 V



Z: 5 V -> 3.3 V



A/B -> PB4/PB5; Z -> PB6 rising–edge EXTI. All original signal polarities preserved.  
Q4–Q6: Diotec 2N2222A E=1/B=2/C=3. Keep collector–to–U3 wires short.  
Check edge shape and transistor turn–off delay at the intended encoder speed.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /M1 – encoder interface/  
File: M1\_Encoder.kicad\_sch

Title: ProtoMate SCARA – M1 – encoder interface

Size: A3

Date: 2026–10–03

Rev: 2.0

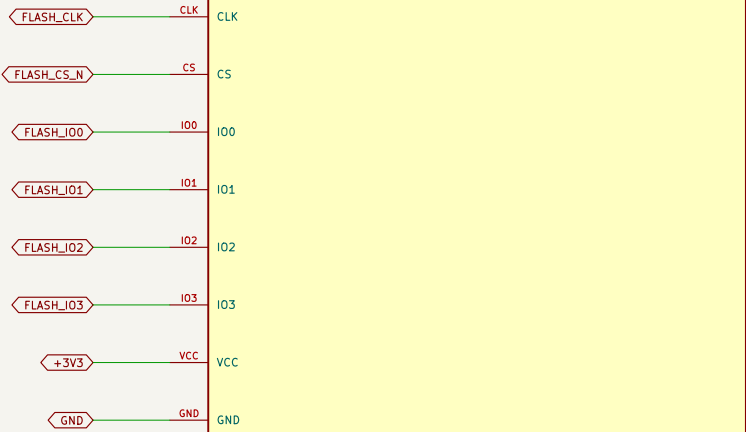
KiCad E.D.A. 10.0.6

Id: 4/7

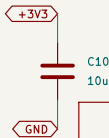
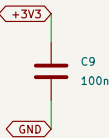
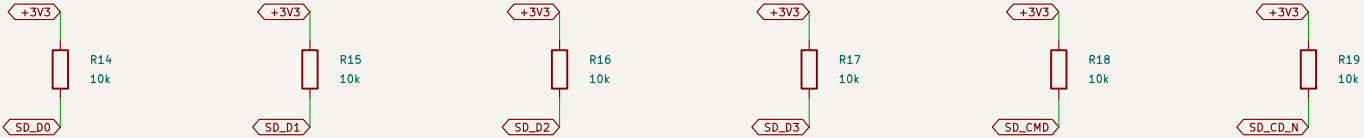
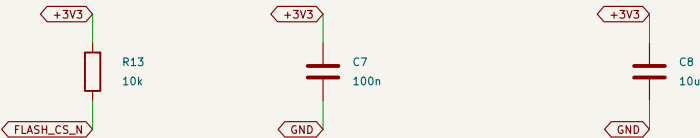
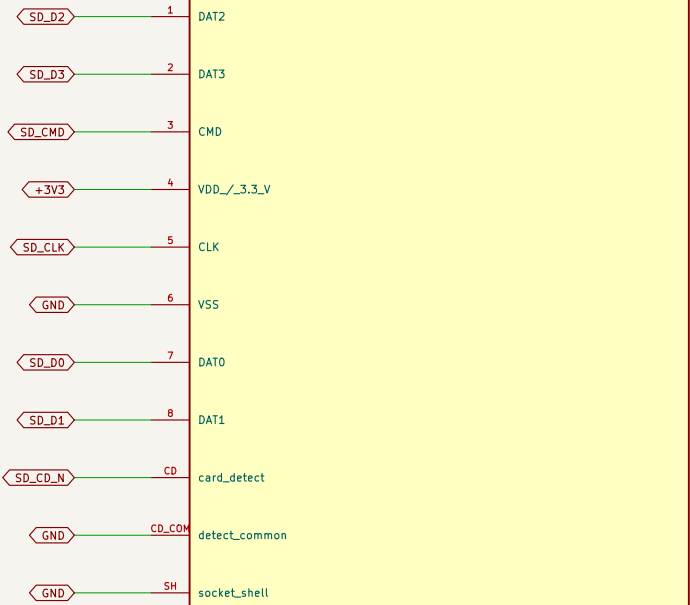
QSPI FLASH AND SD CARD

SD pins 1–8 use standard card–contact numbering.  
CD switch closes to GND when inserted (advisory input).  
Pull–ups on CMD/D0–D3/CD; no pull–up on CLK.  
Existing equivalent module pull–ups/caps need not be duplicated.

A5  
W25Q128 / 3.3 V QSPI



J3  
microSD / 4–bit socket module



3.3 V ONLY  
Use a W25Q128 module exposing IO0–IO3.  
IO2/WP and IO3/HOLD are data in quad mode;  
do not hard–wire them to supply or ground.  
Module contacts are identified by signal name.  
C7/C8 and R13 may already exist on the module;  
fit them if absent. Keep the QSPI harness short.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /QSPI flash and SD card/

File: Storage.kicad\_sch

Title: ProtoMate SCARA – QSPI flash and SD card

Size: A3

Date: 2026–10–02

Rev: 1.0

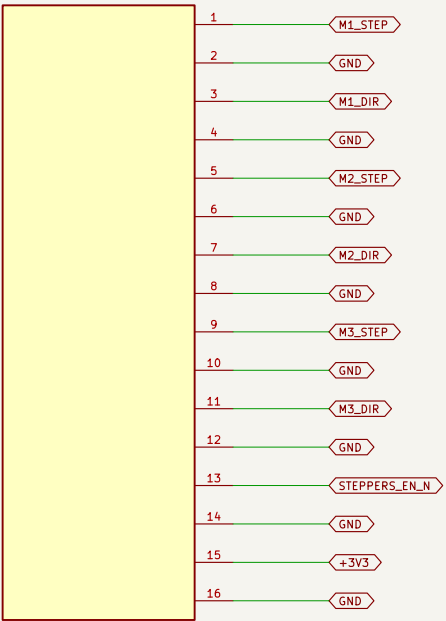
KiCad E.D.A. 10.0.6

Id: 5/7

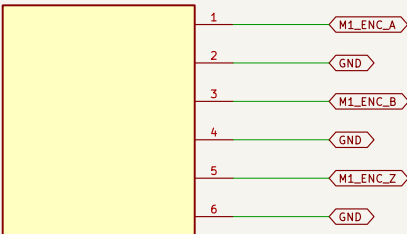
PERFBOARD HEADERS / MODULE MOUNTS

Pin numbers match the layout and Harness.csv. Contacts use the 2.54 mm grid; J103 pairs span 5.08 mm. Mount pins and cable pins are shown.

J101  
Nucleo outputs / 2x08



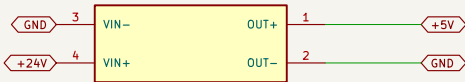
J102  
Nucleo encoder / 2x03



J101/J102/J113: paired even pins are GND.  
J101 pin 15 receives Nucleo 3.3 V.  
J103: U1 mount; OUT+/OUT- feed 5 V/GND.  
J112 is the DM542T 24 V / GND power outlet.

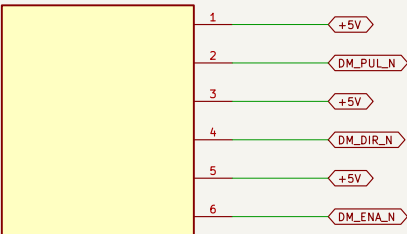
J104: 1 PUL+, 2 PUL-, 3 DIR+, 4 DIR-, 5 ENA+, 6 ENA-.  
J105/J106: 1 VDD, 2 GND, 3 DIR, 4 STEP, 5 MS1, 6 MS2,  
7 DIAG, 8 INDEX, 9 UART shared bus, 10 EN.  
11 = JP1.6 / +24 V; 12 = JP1.5 / GND.  
J107 follows the numbered J2 encoder connector.

J103  
U1 buck direct mount



Pins 3/4: VIN- / GND and VIN+ / 24 V.  
24 V enters U1 through its barrel jack.  
Pins 1/2: regulated OUT+ / OUT-.

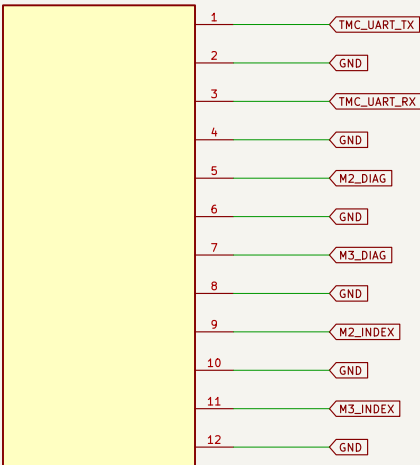
J104  
DM542T / 1x06



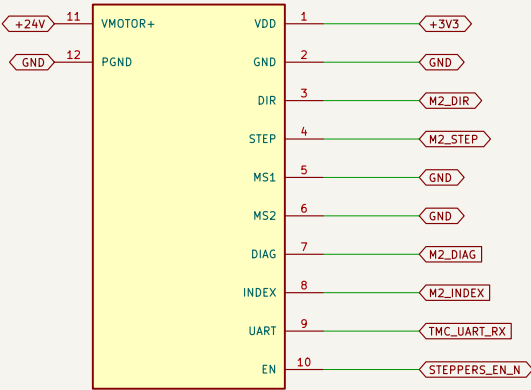
J112  
DM542T power / 5.08 mm



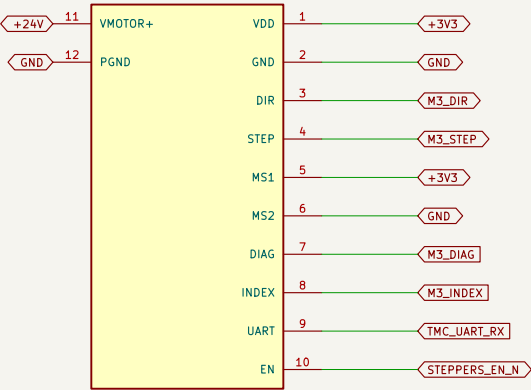
J113  
Nucleo UART / DIAG / INDEX / 2x06



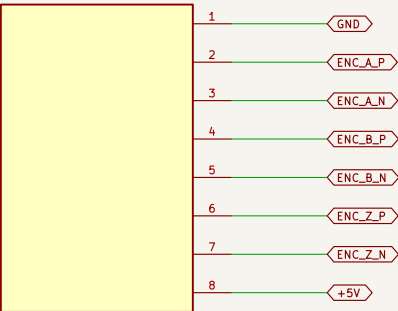
J105  
M2 TMC2209 direct mount



J106  
M3 TMC2209 direct mount



J107  
M1 encoder / 1x08



J101 -> Nucleo CN10: 1->29, 3->26, 5->32, 7->10, 9->34, 11->8, 13->30.  
J103/J105/J106 are direct module mounts, no cable. Other headers use cables. Matching names are one conductor.  
J108-J111 (Nucleo reference inputs and the three NC switches): see the Reference limit switches sheet.

J113 -> CN9: 1->6 (PD5 TX), 3->4 (PD6 RX).  
5->8 (PD4 M2 DIAG), 7->10 (PD3 M3 DIAG).  
9->25 (PDO M2 INDEX), 11->27 (PD1 M3 INDEX).  
Even contacts: GND, TX resistor is R29.  
INDEX: normal electrical cycle; both-edge EXTI; 3.3 V.

J101-J107 and J112-J113 mount on the perfboard; power and signals use global names

ProtoMate Control

Sheet: /Perfboard cable headers/

File: Perfboard\_Connectors.kicad\_sch

Title: ProtoMate SCARA - Perfboard cable headers

Size: A3

Date: 2026-10-03

Rev: 3.0

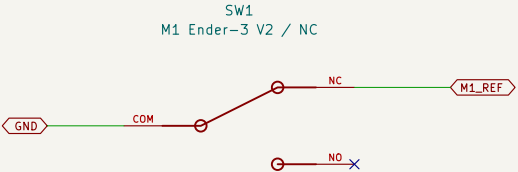
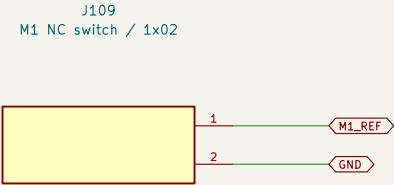
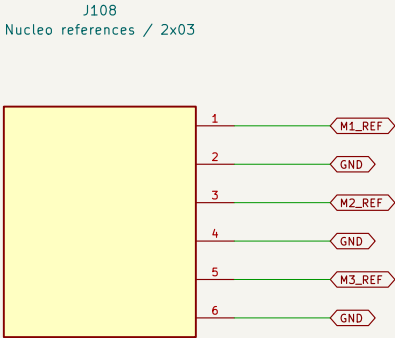
KiCad E.D.A. 10.0.6

Id: 6/7

REFERENCE LIMIT SWITCHES – NORMALLY CLOSED

J108–J111 mount on the perfboard. SW1–SW3 mount on the robot. Global labels connect the cable endpoints.

Switches shown released (COM–NC closed).



J108 -> Nucleo A1  
1 = PE7 / CN10.20 (M1)  
3 = PE8 / CN10.18 (M2)  
5 = PE10 / CN10.24 (M3)  
2, 4, 6 = GND / CN10.22

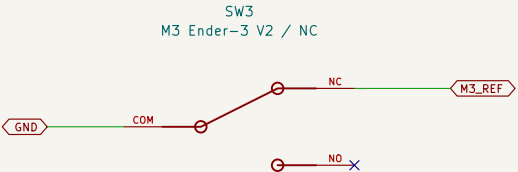
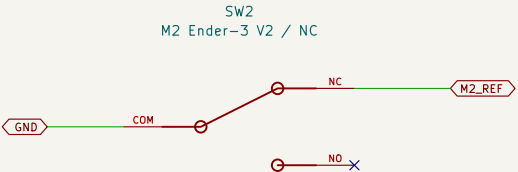
INPUT BEHAVIOR  
Nucleo internal pull–ups to 3.3 V.  
Released: COM–NC closed -> LOW.  
Pressed / unplugged: open -> HIGH.  
Both–edge EXTI: raw, un–debounced inputs.  
Homing and motor–stop logic not yet implemented.

BEFORE WIRING: With the switch disconnected, verify continuity: closed when released, open when pressed.  
COM / NC / NO identify contact functions, NOT the module connector pin order. Do not infer power wiring from V/G/S printing.  
Use the unpopulated mechanical version as a dry contact; no supply connection. Route each REF with its GND return, away from motor wiring.

J109.1 -> verified NC contact  
J109.2 -> verified COM contact  
NO contact stays unconnected.

J110.1 -> verified NC contact  
J110.2 -> verified COM contact  
NO contact stays unconnected.

J111.1 -> verified NC contact  
J111.2 -> verified COM contact  
NO contact stays unconnected.



NC dry contacts | PE7 / PE8 / PE10 | 3.3 V internal pull–ups

ProtoMate Control

Sheet: /Reference limit switches/  
File: Reference\_Switches.kicad\_sch

Title: ProtoMate SCARA – Reference limit switches

Size: A3 Date: 2026–10–03 Rev: 4.0  
KiCad E.D.A. 10.0.6 Id: 7/7